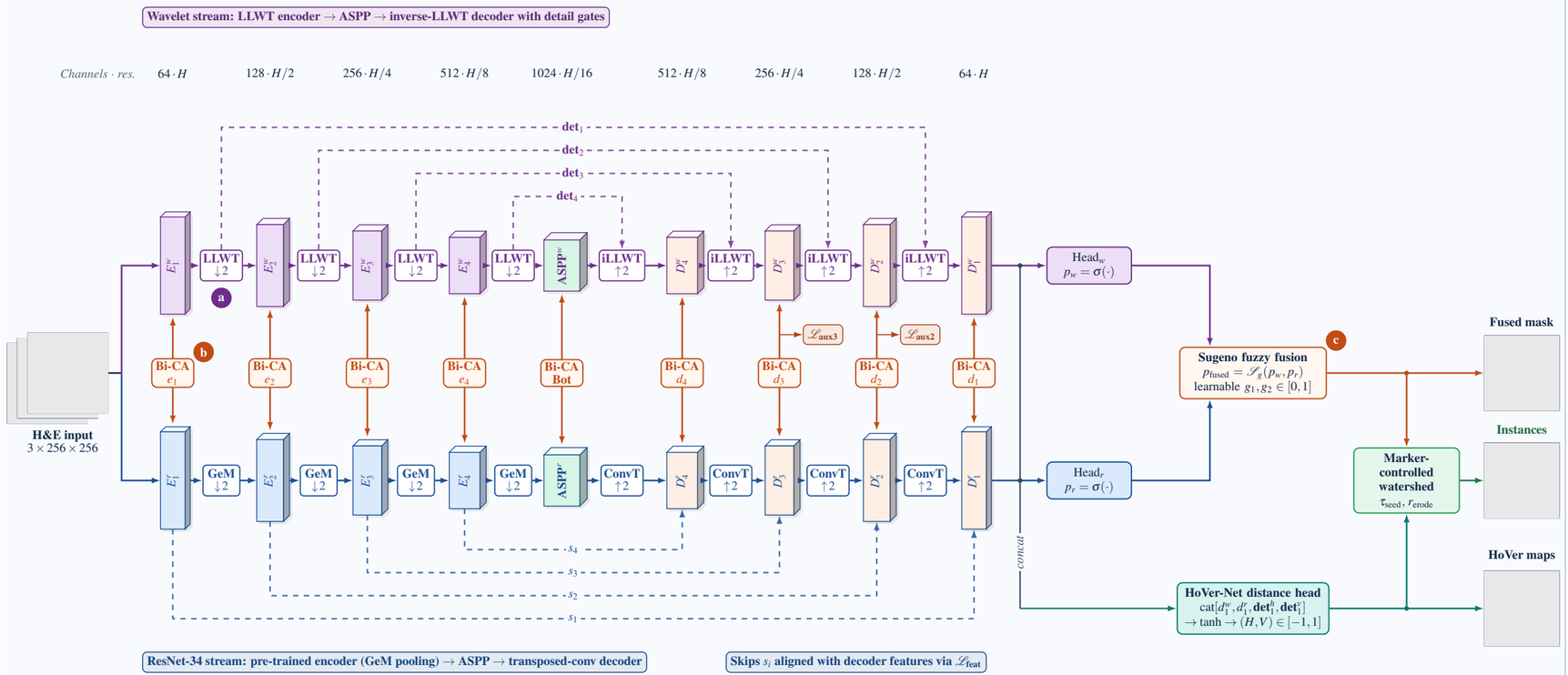
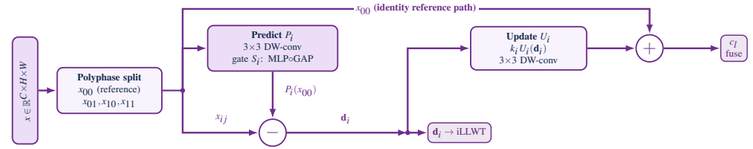
Dual-branch Wavelet–ResNet network for nuclei instance segmentation



a Learnable lifting wavelet transform (LLWT)

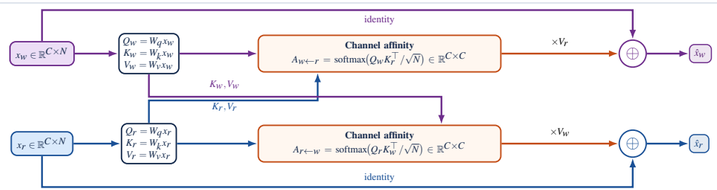
Predict–update lifting with stain-adaptive calibration; inverted exactly by iLLWT



Inverse (decoder): $x_{00} = c_f - \sum_i k_i U_i(\mathbf{d}_i)$ and $x_{ij} = \mathbf{d}_i + P_i(x_{00})$

b Bi-directional channel cross-attention (Bi-CA)

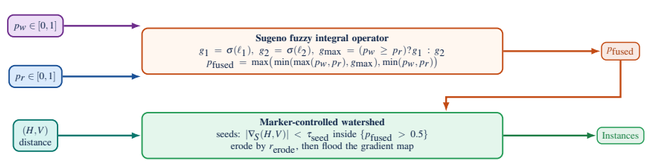
Bridges the two streams at every scale: $E_1 \to E_4$, bottleneck, $D_4 \to D_1$



$\hat{x}_W = x_W + W_{out}(A_{W \leftarrow r} V_r)$; zero-initialised BatchNorm in W_{out} gives an identity mapping at initialization

c Sugeno fuzzy fusion and watershed

Confidence-aware ensembling & marker-controlled instance extraction



p_{fused} is mapped back to logits for Focal-Tversky and BCE supervision during training

Modules Wavelet encoder · ResNet-34 encoder · ASPP bottleneck · Decoder stage · Cross-attention / fusion · HoVer head · Post-processing — *Outline: purple = wavelet, blue = ResNet*

Connections Wavelet path · ResNet path · Detail sub-bands **det**$_i$ to iLLWT · ResNet skip s_i · Cross-attention exchange · Feature concatenation · ⊕ add, ⊖ subtract · **a** – **c** detailed in right side panels